```
import java.util.*;
public class Puzzle {
   static int h(int[][] a) {
      int c=0, k=1;
      for(int[] r:a)
         for(int v:r) {
            if(v!=0 && v!=k) c++;
            k++;
         }
      return c;
   }
   static void print(int[][] a) {
      for(int[] r:a)
         System.out.println(Arrays.toString(r));
      System.out.println();
   }
   public static void main(String[] args) {
      int[][] a={{1,2,3},{4,0,6},{7,5,8}};
      int[][] m={{0,1},{0,-1},{1,0},{-1,0}};
      print(a);
      while(h(a)>0) {
         int x=0,y=0,bx=-1,by=-1,best=999;

         for(int i=0;i<3;i++)
            for(int j=0;j<3;j++)
               if(a[i][j]==0) {x=i;y=j;}
```

```
        for(int[] d:m) {
            int i=x+d[0],j=y+d[1];
            if(i>=0 && i<3 && j>=0 && j<3) {
                int t=a[x][y]; a[x][y]=a[i][j]; a[i][j]=t;
                if(h(a)<best) {best=h(a);bx=i;by=j;}
                t=a[x][y]; a[x][y]=a[i][j]; a[i][j]=t;
            }
        }

        if(bx==-1) break;
        int t=a[x][y]; a[x][y]=a[bx][by]; a[bx][by]=t;
        print(a);
    }
    System.out.println("Goal Reached!");
  }
}
```

3. Theory

A Algorithm:*

A* is an informed search algorithm used to find the shortest path from the starting point to the goal.

It uses two values:

- g(n) – Cost from the starting state to the current state.
- h(n) – Estimated cost from the current state to the goal.
- f(n) – Total estimated cost.

Formula:

\[f(n)=g(n)+h(n) \]

8 Puzzle Problem:

- It consists of 8 numbered tiles and one blank space.
- Tiles can move up, down, left or right.
- The goal is to arrange all tiles in the correct order.
- A* selects the state with the lowest f-score.

Two Lists:

- Open List: Stores states waiting to be explored.
- Closed List: Stores already explored states.

4. Algorithm

1. Start.
2. Define the initial state and goal state.
3. Calculate heuristic value h(n) using misplaced tiles.
4. Set g(n) = 0 for the initial state.
5. Calculate f(n) = g(n) + h(n).
6. Insert the initial state into the priority queue.
7. Select the state with the lowest f-score.
8. If it is the goal state, display the solution and stop.
9. Otherwise, generate new states by moving the blank tile.
10. Calculate g, h and f for each new state.
11. Add new states to the priority queue.
12. Repeat until the goal state is reached.
13. Display the sequence of moves.
14. Stop.

5. Viva Questions and Answers

Q1. What are informed and uninformed search techniques?

- Informed: Uses extra knowledge (heuristic) to find the goal.
- Uninformed: Searches without extra knowledge.

Q2. Difference between informed and uninformed search?

Informed Search	**Uninformed Search**
Uses heuristic	Does not use heuristic
More guided	Less guided
Example: A*	Example: BFS

Q3. What is the evaluation function of A?*

f(n) = g(n) + h(n).

Q4. What are the applications of A algorithm?*

Pathfinding, games, robotics and navigation.

Q5. What is a problem-solving agent?

An agent that finds a sequence of actions to achieve a goal.

Q6. Is A informed or uninformed search?*

A* is an informed search algorithm.

Q7. What is Best First Search?

It selects the next node based on the best evaluation value.

Q8. Where are searching techniques used in AI?

In games, robotics, navigation and problem-solving.

Q9. What are Blind Search algorithms?

Algorithms that search without using heuristic information. Example: BFS and DFS.

Q10. What is a Heuristic?

A heuristic is an estimated value used to reach the goal faster.